Financial Transaction for Risk Prediction

Vivek Rauniyar

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Introduction

The main aim of this project is to explore and visualize patterns within the financial transaction dataset, focusing on the transaction behavior and associated risk levels. As e-banking behaviors have been increasing globally, it becomes extremely important to understand how the factors like user details, their locations, their transaction amount and the timing of those transactions amount contribute to the financial risk. Beside that, there are high chances of increment of the fraudulent transactions, so detecting the anomaly behaviors in the financial market become important for the financial industries to run smoothly and safely. This paper presents five research questions designed to explore relationships between transaction features and risk classification. Each question is supported by visualizations and statistical summaries created in R. The goal is not just to run code, but to interpret the results and understand what factors are meaningful for risk prediction.

Requried Libraries

```r
library(tidyverse)
library(dplyr)
library(ggplot2)
library(scales)
```

Reading the dataset

```r
df <- read.csv("~/Documents/Intro to DS/Project/financial_transactions.csv")
```

Dataset Description

The dataset is extracted from kaggle. We can get the dataset through this link *https://www.kaggle.com/datasets/ziya07/financial-transaction-dataset-for-risk-prediction?utm_source=chatgpt.com* It contains the various columns like transaction ID, Sender ID, receiver ID, Timestamp, Transaction Type, Device ID, IP Address, Location, Account Balance, and based upon these inputs, we have to predict whether the transaction is fraudulent or not. The dataset has 10000 rows and 15 columns. Data preprocessing should be done to clean the data, since we have many null values in our dataset and there are some columns in our dataset which hold no values in it. Those unnecessary columns are simply removed after the data preprocessing. Each row represets a single transaction and includes attributes such as:

Type: Whether the transaction in a Deposit, Transfer, or Withdrawal

Amount: The value of the transaction in dollars

Timestamp: The data and time the transaction occurred

Location: The U.S. state where the transaction was processed

Suspicious.Activity.Flag: Indicated whether the transaction was flagged as Suspicious (1) or Not Suspicious(0). It is important because it is often used as a label for machine learning risk classification or for

identifying patterns in suspicious behaviour.

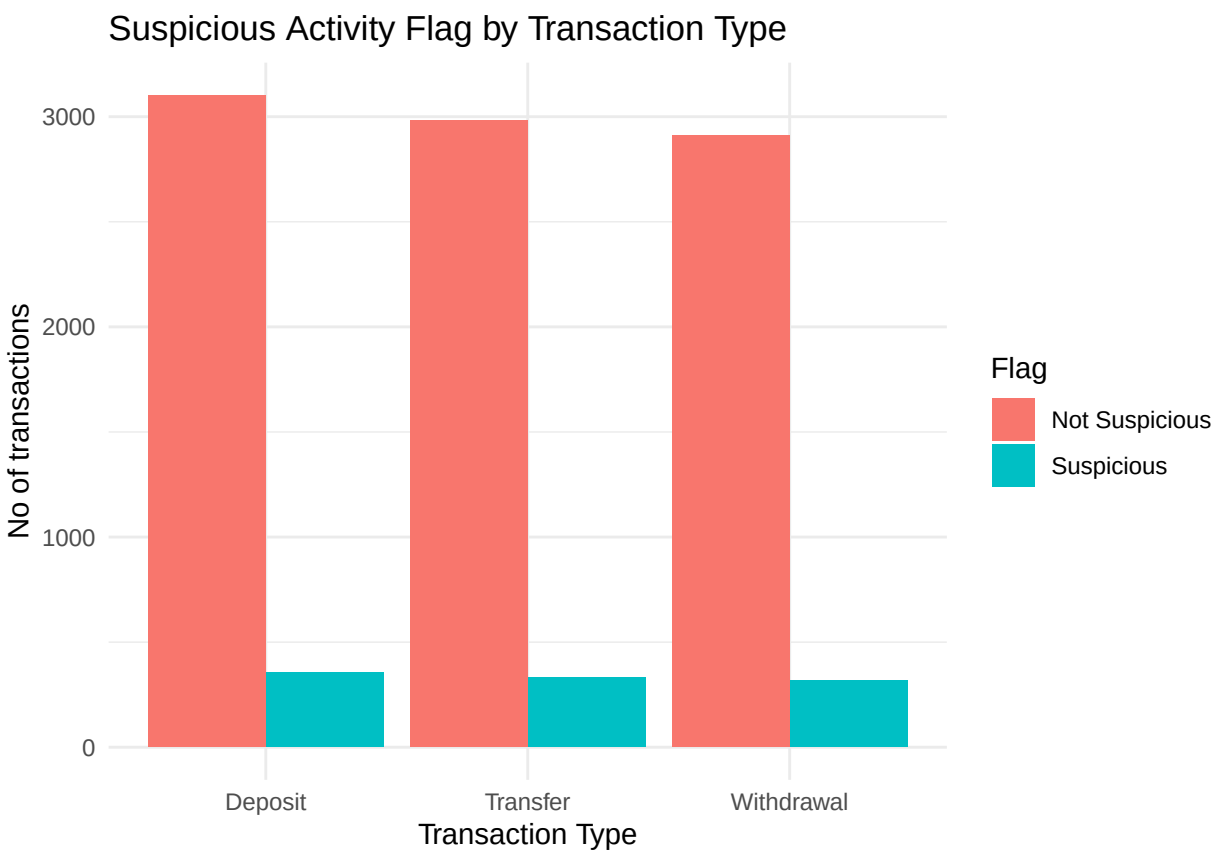
We have few more colums like User.Device.Recognition, Known.Threat.Flag, Login.Location.Match, Spending.Pattern.Deviation. I am not doing any analysis on these columns so I just ignore these columns for this project.

Tools and Methodology

I used R programming language to read the data, to preprocess it and using the library called ggplot2, I have done the data visualization. All the analysis have been performed in RStudio. I have found out the relation between each input and the output part, showing the correlation between them and find out how heavily one influences the other. I have also created the graphs like barplot, histogram, and bargraphs to find the relation between the features.

```
# Question 1:   Are certain transaction types (e.g., deposit, transfer, and withdrawal) more
# likely to be classified as high-risk classification?
```

```
# Dodge helps to draw the bars for each flag side by side (grouped) not stacked.
ggplot(df, aes(x = Type, fill = Suspicious.Activity.Flag)) +
  geom_bar(position = "dodge") +
  labs(
    title = "Suspicious Activity Flag by Transaction Type",
    x = "Transaction Type",
    y = 'No of transactions',
    fill = "Flag"
  ) +
  theme_minimal()
```



The bar chart displays the number of suspicious and non-suspicious transactions across the three transaction

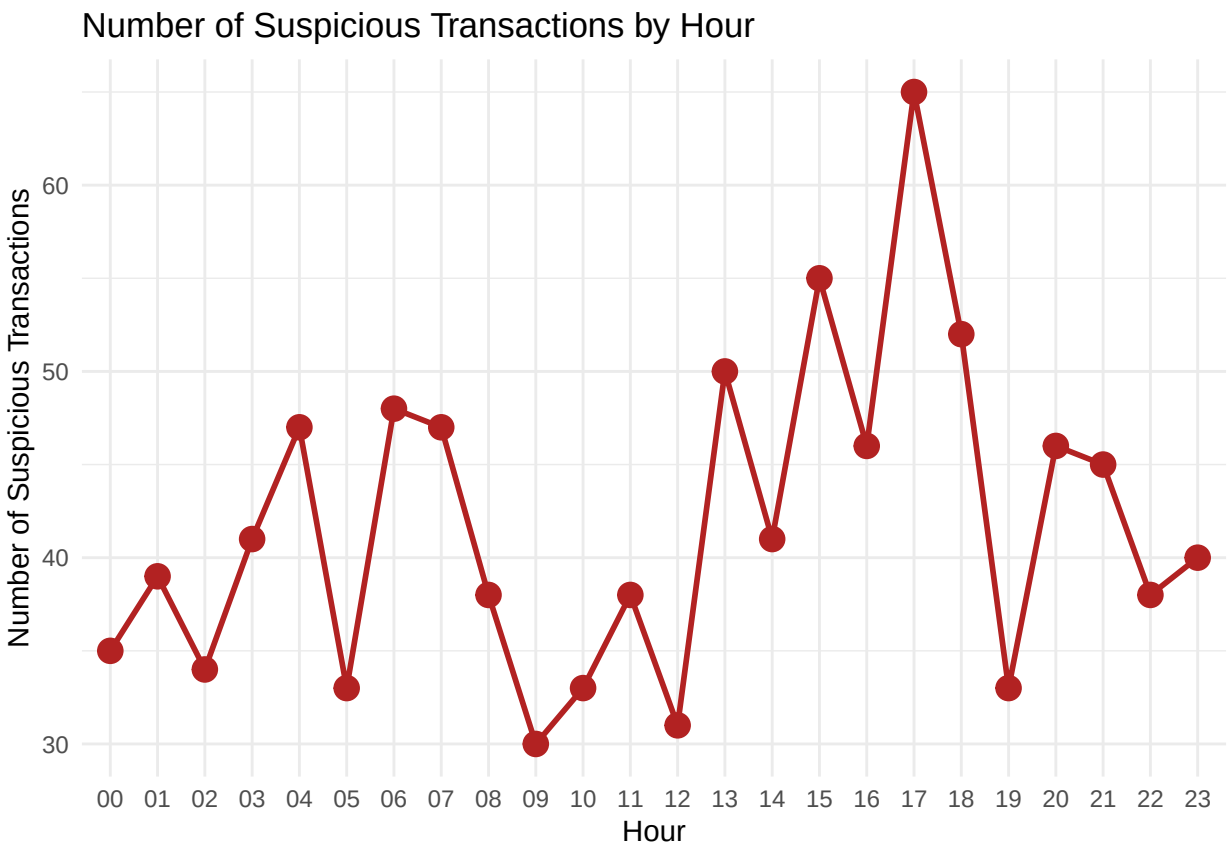
types: Deposit, Transfer, and Withdrawal. Across all transaction types, the majority of transactions are classified as Not Suspicious, represented by the taller red bars. In comparison, suspicious transactions (shown in blue) account for a much smaller proportion of the total volume. Although all three types exhibit some level of suspicious activity, Deposits appear to have the highest count of suspicious transactions, followed closely by Transfers and Withdrawals. This suggests that while suspicious behavior occurs across all transaction types, deposits may be slightly more prone to triggering the suspicious activity flag in this dataset. However, the overall distribution indicates that suspicious transactions remain a small fraction relative to legitimate ones.

```r
# Question 2:   Does the timing of a transaction influence its risk category?
```

```r
# Convert Timestamp to POSIXct
df$Timestamp <- as.POSIXct(df$Timestamp)

# Extract hour (0-23) as a categorical variable
df$Hour <- format(df$Timestamp, "%H")

# Summarize frequency by hour and risk flag
df_suspicious <- df %>%
  filter(Suspicious.Activity.Flag=="Suspicious")

suspicious_by_hour <- df_suspicious %>%
  group_by(Hour)%>%
  summarise(
    suspicious_count = n()
  )%>%
  arrange(Hour)

suspicious_by_hour
```

```
## # A tibble: 24 x 2
##    Hour  suspicious_count
##    <chr>            <int>
##  1 00                  35
##  2 01                  39
##  3 02                  34
##  4 03                  41
##  5 04                  47
##  6 05                  33
##  7 06                  48
##  8 07                  47
##  9 08                  38
## 10 09                  30
## # i 14 more rows
```

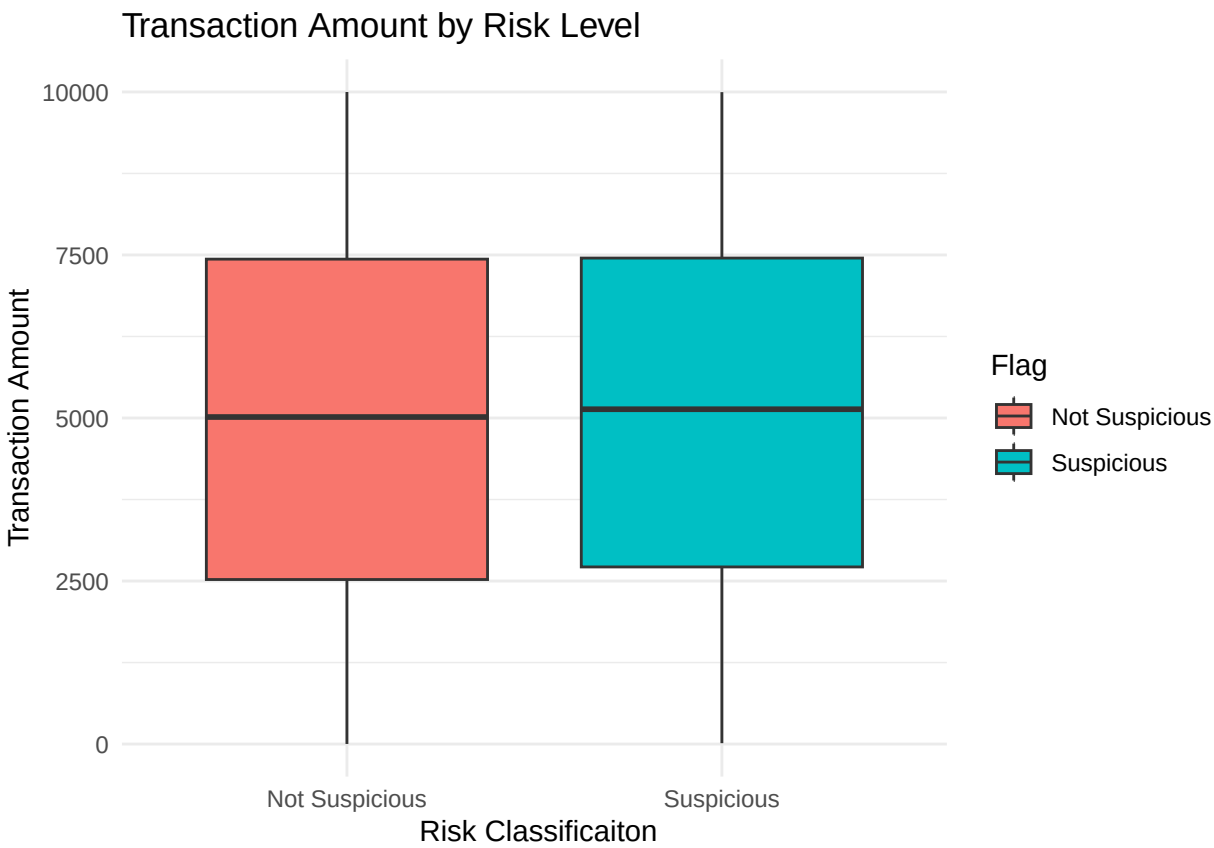
```r
ggplot(suspicious_by_hour,
       aes(x = Hour, y = suspicious_count, group = 1)) +
  geom_line(color = "firebrick", linewidth = 1) +
  geom_point(color = "firebrick", size = 4) +
  labs(
    title = "Number of Suspicious Transactions by Hour",
    x = "Hour",
    y = "Number of Suspicious Transactions"
  ) +
  theme_minimal()
```



This graph shows the number of suspicious transactions for each hour of the day. The number of suspicious transactions varies throughout the day rather than remaining constant. There are certain hours where suspicious transactions are higher, especially in the late afternoon and early evening hours. In contrast, fewer suspicious transactions occur during the early morning. This indicates that suspicious transactions do not occur randomly and are more common at certain times of the day. Overall, the time of a transaction appears to have some influence on when suspicious activity is more likely to occur.

```
# Question 3: How is the transaction amount related to the risk level? For example, are very large or
# very small transactions more often flagged as risky?
```

```
ggplot(df, aes(x = Suspicious.Activity.Flag, y = Amount, fill = Suspicious.Activity.Flag))+
  geom_boxplot()+
  labs(
    title = "Transaction Amount by Risk Level",
    x = " Risk Classificaiton",
    y = "Transaction Amount",
    fill = "Flag"
  ) +
  theme_minimal()
```



This graph compares how much money is involved in suspicious and non-suspicious transactions. Both types of transactions have very similar amounts. The middle lines in the boxes are almost the same, which means the typical transaction amount is nearly equal for both groups. Suspicious transactions do show a little more variation, but not enough to say that suspicious activity always involves bigger or smaller amounts. Thus, the amount of money does not seem to decide whether a transaction is suspicious. People can make large or small transactions that are both safe or suspicious.

```
df$Amount <- as.numeric(df$Amount)
df$Suspicious.Activity.Flag <- factor(df$Suspicious.Activity.Flag)
table(df$Suspicious.Activity.Flag)
```

```
##
## Not Suspicious     Suspicious
##           8995           1005
```

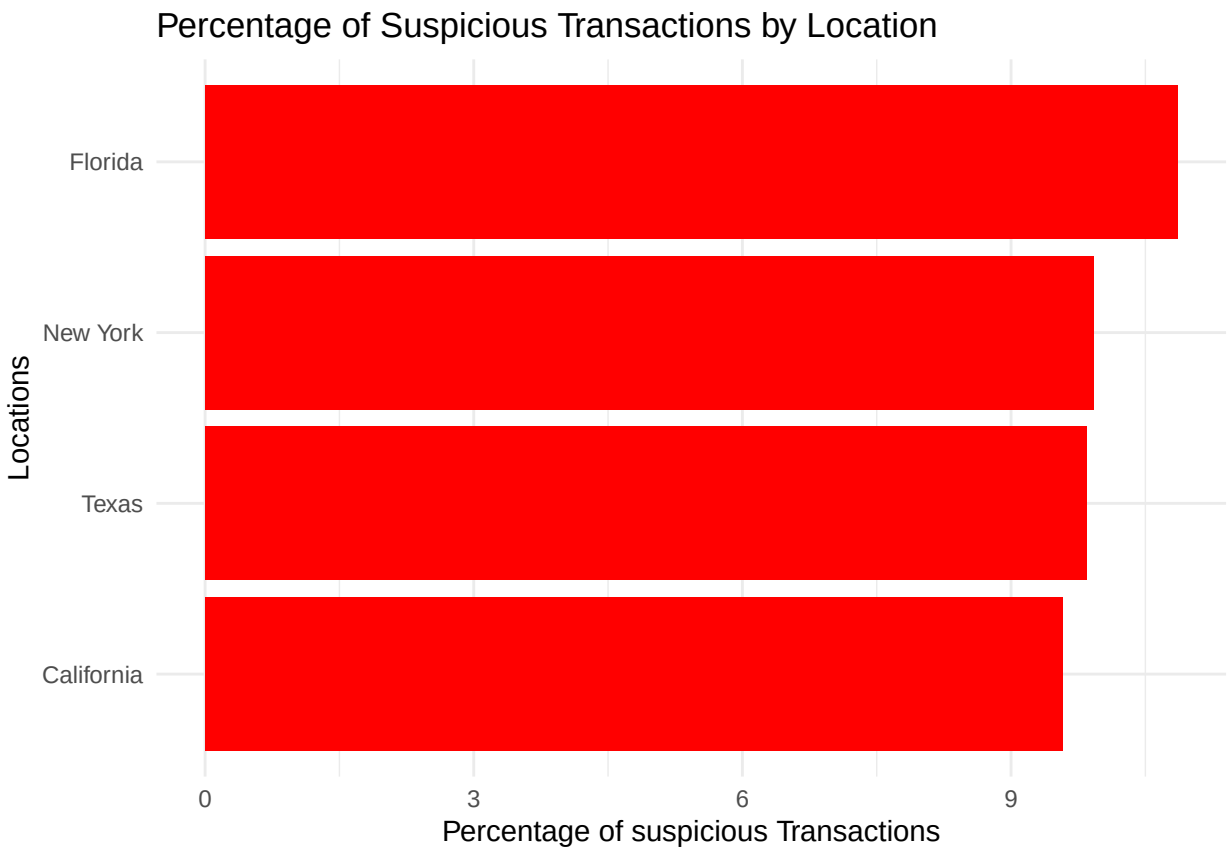
```
t_test_result <- t.test(Amount ~ Suspicious.Activity.Flag, data = df)
t_test_result
```

```
##
##  Welch Two Sample t-test
##
## data:  Amount by Suspicious.Activity.Flag
## t = -0.58833, df = 1245.2, p-value = 0.5564
## alternative hypothesis: true difference in means between group Not Suspicious and group Suspicious is
## 95 percent confidence interval:
##  -240.8261  129.7085
## sample estimates:
## mean in group Not Suspicious     mean in group Suspicious
##                     4996.938                     5052.497
```

Conclusion from t-test:A t-test was used to compare the average transaction amounts for suspicious and non-suspicious transactions. The results show no statistically significant difference between the two groups **(p = 0.56)**. The average amounts are almost the same for both types of transactions. This means that transaction amount by itself does not strongly affect whether a transaction is marked as suspicious.

```
# Question 4: Do transaction locations show patterns, such as certain regions or
# areas being more prone to high-risk transactions?
```

```
# Summarize suspicious percentage by location
location_suspicious <- df %>%
  group_by(Location) %>%
  summarise(
    suspicious_percent = mean(Suspicious.Activity.Flag == "Suspicious") * 100,
    count = n()
  ) %>%
  arrange(desc(suspicious_percent))

# Plot only suspicious percentage by location
ggplot(location_suspicious,
       aes(x = reorder(Location, suspicious_percent),
           y = suspicious_percent)) +
  geom_col(fill = "red") +
  coord_flip() +
  labs(
    title = "Percentage of Suspicious Transactions by Location",
    x = "Locations",
    y = "Percentage of suspicious Transactions"
  ) +
  theme_minimal()
```



This graph shows the percentage of suspicious transactions in each location, sorted from the highest to the lowest. By focusing only on the proportion of suspicious activity, the chart makes it easy to compare risk levels across states. We can clearly see that some locations, such as Florida, have a higher percentage of suspicious transactions compared to others like California and New York. This indicates that geographic patterns do exist: certain states appear more prone to risky transactions. However, the differences are not extremely large, suggesting that location may play a role in suspicious behavior, but it is not the only factor influencing fraud risk.

```
df$Location <- factor(df$Location)
df$Suspicious.Activity.Flag <- factor(df$Suspicious.Activity.Flag)

location_risk_table <- table(df$Location, df$Suspicious.Activity.Flag)

chi_location <- chisq.test(location_risk_table)
chi_location
```

```
##
##  Pearson's Chi-squared test
##
## data:  location_risk_table
## X-squared = 2.5893, df = 3, p-value = 0.4594
```

Conclusion from Chi-square test: Since the **p-value** is much greater than **0.05** , it concludes that there is no statistically significant association between location and risk classificaiton. Any differences that is seen across locations are likely due to random variation, not a real underlying pattern.

```
# Question 5: What is the average transaction amount for each risk category,
# and how do these distributions compare?
```

```r
df %>%
  group_by(Suspicious.Activity.Flag) %>%
  summarise(
    Average_Amount = mean(Amount),
    Median_Amount = median(Amount),
    Count = n()
  )
```

```
## # A tibble: 2 x 4
##   Suspicious.Activity.Flag Average_Amount Median_Amount Count
##   <fct>                              <dbl>         <dbl> <int>
## 1 Not Suspicious                     4997.         5015.  8995
## 2 Suspicious                         5052.         5134.  1005
```

```r
ggplot(df, aes(x = Suspicious.Activity.Flag, y = Amount, fill = Suspicious.Activity.Flag))+
  geom_boxplot()+
  labs(
    title = "Distribution of Transaction Amounts by Risk Category",
    x = "Risk Category",
    y = "Transaction Amount",
    fill ='Flag'
  ) +
  theme_minimal()
```

The average amount of money in suspicious transactions is about $5,052, and the average for non-suspicious transactions is about $4,997. These values are almost the same, which means suspicious transactions are not based on unusually high or low amounts. The median amounts are also very close. However, even though the average amounts are similar, suspicious transactions make up only about 10% of all transactions (1,005 out

of 9,995). This shows that most transactions are safe, and only a small portion are flagged as suspicious. So, the amount of money does not seem to be the main reason a transaction becomes suspicious, the system must be using other factors to decide which ones to flag.

Results

After performing an exploratory data analysis on the transaction dataset, several patterns emerged that help us understand how risk is assessed in financial systems. First, the analysis of transaction types showed that **Deposits, Transfers, and Withdrawals all contain a small share of suspicious transactions**, but Deposits had the highest count. However, most transactions in every category were not suspicious, indicating that transaction type alone is not a strong predictor of risk. Next, studying the timing of transactions revealed that **suspicious transactions occur throughout the day**, but tend to appear slightly more often during **early morning and late afternoon hours**. This suggests that time may contribute to suspicious transaction behavior, although it is not the primary determining factor. Comparing transaction amounts for each risk category showed that **both suspicious and non-suspicious transactions share nearly identical average and median values**. This means that the amount of money being transferred is not a major indicator of risk. People can make large or small transactions that may or may not be suspicious; therefore, the system relies on other behavioral factors to identify unusual activity. When examining transaction locations, we found that **some states, such as Florida, have a slightly higher percentage of suspicious transactions compared to others like New York or Texas**. Although these differences are small, they suggest that location may offer useful context in identifying potential fraud patterns. Overall, the results show that while some features such as transaction type, timing, and location provide minor clues about risk, **no single factor is enough to classify a transaction as suspicious**. Instead, risk prediction appears to be influenced by a combination of behavioral patterns, which aligns with how real financial fraud detection systems operate.

Conclusion

This project explored five different aspects of financial transactions to determine whether they contribute to classifying a transaction as suspicious. The analysis showed that **most transactions are legitimate**, and only a small portion are marked as suspicious. Factors such as transaction type, time of day, and location do show slight patterns, but they are not strong enough to predict risk on their own. The most important finding is that **transaction amount does not significantly influence risk classification**, as both suspicious and non-suspicious transactions have almost identical average amounts. This confirms that modern financial systems do not rely on a single feature to detect fraud. Instead, they use multiple layers of information—such as device recognition, spending behavior, and known threat indicators—to identify unusual or risky activity. In summary, this study highlights the importance of analyzing multiple variables together rather than relying on one factor. Understanding these patterns can help improve fraud detection systems, reduce false alerts, and protect users from financial threats. As financial transactions continue to grow in volume and complexity, combining behavioral, geography, and temporal data will remain essential for accurate risk prediction.